

Equation Sheet for ASEN 3728 Final Exam

Fall 2025

Rotation Matrices:

$$\mathbf{R}_E^B(\phi, \theta, \psi) = \begin{pmatrix} c_\theta c_\psi & c_\theta s_\psi & -s_\theta \\ s_\phi s_\theta c_\psi - c_\phi s_\psi & s_\phi s_\theta s_\psi + c_\phi c_\psi & s_\phi c_\theta \\ c_\phi s_\theta c_\psi + s_\phi s_\psi & c_\phi s_\theta s_\psi - s_\phi c_\psi & c_\phi c_\theta \end{pmatrix}$$

$$\mathcal{R}_b^s(\alpha) = \begin{pmatrix} \cos \alpha & 0 & \sin \alpha \\ 0 & 1 & 0 \\ -\sin \alpha & 0 & \cos \alpha \end{pmatrix} \quad \mathcal{R}_s^w(\beta) = \begin{pmatrix} \cos \beta & \sin \beta & 0 \\ -\sin \beta & \cos \beta & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Newton-Euler equations of motion for conventional aircraft:

$$\begin{pmatrix} \dot{x}_E \\ \dot{y}_E \\ \dot{z}_E \end{pmatrix} = \begin{pmatrix} c_\theta c_\psi & s_\phi s_\theta c_\psi - c_\phi s_\psi & c_\phi s_\theta c_\psi + s_\phi s_\psi \\ c_\theta s_\psi & s_\phi s_\theta s_\psi + c_\phi c_\psi & c_\phi s_\theta s_\psi - s_\phi c_\psi \\ -s_\theta & s_\phi c_\theta & c_\phi c_\theta \end{pmatrix} \begin{pmatrix} u^E \\ v^E \\ w^E \end{pmatrix}$$

$$\begin{pmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{pmatrix} = \begin{pmatrix} 1 & \sin \phi \tan \theta & \cos \phi \tan \theta \\ 0 & \cos \phi & -\sin \phi \\ 0 & \sin \phi \sec \theta & \cos \phi \sec \theta \end{pmatrix} \begin{pmatrix} p \\ q \\ r \end{pmatrix}$$

$$\begin{pmatrix} \dot{u}^E \\ \dot{v}^E \\ \dot{w}^E \end{pmatrix} = \begin{pmatrix} rv^E - qw^E \\ pw^E - ru^E \\ qu^E - pv^E \end{pmatrix} + g \begin{pmatrix} -\sin \theta \\ \cos \theta \sin \phi \\ \cos \theta \cos \phi \end{pmatrix} + \frac{1}{m} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix}$$

$$\begin{pmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{pmatrix} = \begin{pmatrix} \Gamma_1 pq - \Gamma_2 qr \\ \Gamma_5 pr - \Gamma_6 (p^2 - r^2) \\ \Gamma_7 pq - \Gamma_8 qr \end{pmatrix} + \begin{pmatrix} \Gamma_3 L + \Gamma_4 N \\ \frac{1}{I_y} M \\ \Gamma_4 L + \Gamma_8 N \end{pmatrix}$$

$$\Gamma_1 = \frac{I_{xz}(I_x - I_y + I_z)}{\Gamma}$$

$$\Gamma_2 = \frac{I_z(I_z - I_y) + I_{xz}^2}{\Gamma}$$

$$\Gamma_3 = \frac{I_z}{\Gamma}$$

$$\Gamma_4 = \frac{I_{xz}}{\Gamma}$$

$$\Gamma_5 = \frac{I_z - I_x}{I_y}$$

$$\Gamma_6 = \frac{I_{xz}}{I_y}$$

$$\Gamma_7 = \frac{I_x(I_x - I_y) + I_{xz}^2}{\Gamma}$$

$$\Gamma_8 = \frac{I_x}{\Gamma}$$

$$\Gamma = I_x I_z - I_{xz}^2$$

Quadcopter equations of motion (dynamics only)

$$\begin{pmatrix} \dot{u}^E \\ \dot{v}^E \\ \dot{w}^E \end{pmatrix} = \begin{pmatrix} rv^E - qw^E \\ pw^E - ru^E \\ qu^E - pv^E \end{pmatrix} + g \begin{pmatrix} -\sin \theta \\ \cos \theta \sin \phi \\ \cos \theta \cos \phi \end{pmatrix} + \frac{1}{m} \begin{pmatrix} X \\ Y \\ Z \end{pmatrix} + \frac{1}{m} \begin{pmatrix} 0 \\ 0 \\ Z_c \end{pmatrix}$$

$$\begin{pmatrix} \dot{p} \\ \dot{q} \\ \dot{r} \end{pmatrix} = \begin{pmatrix} \frac{I_y - I_z}{I_x} qr \\ \frac{I_z - I_x}{I_y} pr \\ \frac{I_x - I_y}{I_z} pq \end{pmatrix} + \begin{pmatrix} \frac{1}{I_x} L \\ \frac{1}{I_y} M \\ \frac{1}{I_z} N \end{pmatrix} + \begin{pmatrix} \frac{1}{I_x} L_c \\ \frac{1}{I_y} M_c \\ \frac{1}{I_z} N_c \end{pmatrix}$$

Linearized quadcopter equations of motion for steady hover:

$$\begin{pmatrix} \Delta \dot{x}_E \\ \Delta \dot{u} \\ \Delta \dot{\theta} \\ \Delta \dot{q} \end{pmatrix} = \begin{pmatrix} \Delta u \\ -g \Delta \theta \\ \Delta q \\ \frac{1}{I_y} \Delta M_c \end{pmatrix} \quad \begin{pmatrix} \Delta \dot{y}_E \\ \Delta \dot{v} \\ \Delta \dot{\phi} \\ \Delta \dot{p} \end{pmatrix} = \begin{pmatrix} \Delta v \\ g \Delta \phi \\ \Delta p \\ \frac{1}{I_x} \Delta L_c \end{pmatrix}$$

$$\begin{pmatrix} \Delta \dot{z}_E \\ \Delta \dot{w} \end{pmatrix} = \begin{pmatrix} \Delta w \\ \frac{1}{m} \Delta Z_c \end{pmatrix} \quad \begin{pmatrix} \Delta \dot{\psi} \\ \Delta \dot{r} \end{pmatrix} = \begin{pmatrix} \Delta r \\ \frac{1}{I_z} \Delta N_c \end{pmatrix}$$

Other general equations:

$$\frac{d}{dt} \mathbf{p}_B = \dot{\mathbf{p}}_B + (\boldsymbol{\omega}_B \times \mathbf{p}_B) = \dot{\mathbf{p}}_B + \tilde{\boldsymbol{\omega}}_B \mathbf{p}_B \quad \tilde{\boldsymbol{\omega}}_B = \begin{pmatrix} 0 & -r & q \\ r & 0 & -p \\ -q & p & 0 \end{pmatrix}$$

$$\alpha = \tan^{-1} \frac{w}{u} \quad \beta = \sin^{-1} \frac{v}{V}$$

$$V = |\mathbf{V}_B| = \sqrt{u^2 + v^2 + w^2}$$

$$\begin{pmatrix} u \\ v \\ w \end{pmatrix} = V_a \begin{pmatrix} \cos \alpha \cos \beta \\ \sin \beta \\ \sin \alpha \cos \beta \end{pmatrix}$$

Nondimensionalization (Table 4.1 in Etkin and Reid):

Dimensional Quantity	Divisor (General Case)	Divisor (Small-Disturbance Case)	Nondimensional Quantity
X, Y, Z	$\frac{1}{2}\rho V^2 S$	$\frac{1}{2}\rho V^2 S$	C_x, C_y, C_z
mg	$\frac{1}{2}\rho V^2 S$	$\frac{1}{2}\rho V^2 S$	C_W
M	$\frac{1}{2}\rho V^2 S \bar{c}$	$\frac{1}{2}\rho V^2 S \bar{c}$	C_m
L, N	$\frac{1}{2}\rho V^2 S b$	$\frac{1}{2}\rho V^2 S b$	C_l, C_n
u, v, w	V	u_0	$\hat{u}, \hat{v}, \hat{w} (1)$
$\dot{\alpha}, q$	$2V/\bar{c}$	$2u_0/\bar{c}$	$\hat{\alpha}, \hat{q}$
$\dot{\beta}, p, r$	$2V/b$	$2u_0/b$	$\hat{\beta}, \hat{p}, \hat{r}$
m	$\rho S \bar{c}/2$	$\rho S \bar{c}/2$	μ
I_y	$\rho S (\bar{c}/2)^3$	$\rho S (\bar{c}/2)^3$	$\hat{I}$
I_x, I_z, I_{zx}	$\rho S (b/2)^3$	$\rho S (b/2)^3$	$\hat{I}_x, \hat{I}_z, \hat{I}_{zx}$

Second Order Dynamical Systems Analysis:

$$\lambda = \sigma \pm jd = -\xi\omega_n \pm j\omega_d$$

$$\omega_n = \sqrt{\sigma^2 + d^2}, \quad \omega_d = \omega_n \sqrt{1 - \xi^2}, \quad \tau = \frac{-1}{\sigma}$$

Longitudinal trim and static stability analysis:

$$\begin{bmatrix} C_{L_\alpha} & C_{L_{\delta_e}} \\ C_{m_\alpha} & C_{m_{\delta_e}} \end{bmatrix} \begin{bmatrix} \alpha_{trim} \\ \delta_{e_{trim}} \end{bmatrix} = \begin{bmatrix} C_{L_{trim}} \\ -C_{m_0} \end{bmatrix} \quad h_n = h_{n_{wb}} - \frac{1}{C_{L_\alpha}} \left(\frac{\partial C_{m_{acwb}}}{\partial \alpha} - \bar{V}_H \frac{\partial C_{L_t}}{\partial \alpha} + \frac{\partial C_{m_p}}{\partial \alpha} \right)$$

$$C_{m_\alpha} = C_{L_\alpha} (h - h_n) \quad \text{static margin} = K_n = h_n - h \quad \bar{V}_H = \frac{\bar{l}_t}{\bar{c}} \frac{S_t}{S}$$

Linearized longitudinal fixed-wing aircraft equations of motion for steady level flight:

$$\dot{\mathbf{x}}_{lon} = \mathbf{A}_{lon} \mathbf{x}_{lon} + \mathbf{c}_{lon}$$

$$\mathbf{x}_{lon} = \begin{pmatrix} \Delta u \\ \Delta w \\ \Delta q \\ \Delta \theta \end{pmatrix} \quad \mathbf{c}_{lon} = \begin{pmatrix} \frac{\Delta X_c}{m} \\ \frac{\Delta Z_c}{m - Z_{\dot{w}}} \\ \frac{\Delta M_c}{I_y} + \frac{M_{\dot{w}}}{I_y} \frac{\Delta Z_c}{(m - Z_{\dot{w}})} \\ 0 \end{pmatrix}$$

$$\mathbf{A}_{lon} = \begin{pmatrix} \frac{X_u}{m} & \frac{X_w}{m} & 0 & -g \cos \theta_0 \\ \frac{Z_u}{m - Z_{\dot{w}}} & \frac{Z_w}{m - Z_{\dot{w}}} & \frac{Z_q + mu_0}{m - Z_{\dot{w}}} & \frac{-mg \sin \theta_0}{m - Z_{\dot{w}}} \\ \frac{1}{I_y} \left[M_u + \frac{M_{\dot{w}} Z_u}{m - Z_{\dot{w}}} \right] & \frac{1}{I_y} \left[M_w + \frac{M_{\dot{w}} Z_w}{m - Z_{\dot{w}}} \right] & \frac{1}{I_y} \left[M_q + \frac{M_{\dot{w}} (Z_q + mu_0)}{m - Z_{\dot{w}}} \right] & \frac{-M_{\dot{w}} mg \sin \theta_0}{I_y (m - Z_{\dot{w}})} \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

$$\dot{\mathbf{x}}_{lon} = \mathbf{A}_{lon} \mathbf{x}_{lon} + \mathbf{B}_{lon} \mathbf{u}_{lon}$$

$$\mathbf{B}_{lon} = \begin{pmatrix} \frac{X_{\delta_e}}{m} & \frac{X_{\delta_t}}{m} \\ \frac{Z_{\delta_e}}{m - Z_{\dot{w}}} & \frac{Z_{\delta_t}}{m - Z_{\dot{w}}} \\ \frac{M_{\delta_e}}{I_y} + \frac{M_{\dot{w}}}{I_y} \frac{Z_{\delta_e}}{(m - Z_{\dot{w}})} & \frac{M_{\delta_t}}{I_y} + \frac{M_{\dot{w}}}{I_y} \frac{Z_{\delta_t}}{(m - Z_{\dot{w}})} \\ 0 & 0 \end{pmatrix} \quad \mathbf{u}_{lon} = \begin{pmatrix} \Delta \delta_e \\ \Delta \delta_t \end{pmatrix}$$

Short period and phugoid approximations:

$$\begin{pmatrix} \Delta \dot{w} \\ \Delta \dot{q} \end{pmatrix} = \begin{pmatrix} \frac{Z_w}{m} & u_0 \\ \frac{1}{I_y} \left[M_w + \frac{M_{\dot{w}} Z_w}{m} \right] & \frac{1}{I_y} [M_q + M_{\dot{w}} u_0] \end{pmatrix} \begin{pmatrix} \Delta w \\ \Delta q \end{pmatrix}$$

$$\begin{pmatrix} \Delta \dot{u} \\ \Delta \dot{\theta} \end{pmatrix} = \begin{pmatrix} X_u/m & -g \\ -\frac{Z_u}{mu_0} & 0 \end{pmatrix} \begin{pmatrix} \Delta u \\ \Delta \theta \end{pmatrix}$$

Linearized lateral fixed-wing aircraft equations of motion for steady level flight:

$$\dot{\mathbf{x}}_{lat} = \mathbf{A}_{lat} \mathbf{x}_{lat} + \mathbf{B}_{lat} \mathbf{u}_{lat}$$

$$\mathbf{A}_{lat} = \begin{pmatrix} \frac{Y_v}{m} & \frac{Y_p}{m} & \left(\frac{Y_r}{m} - u_0 \right) & g \cos \theta_0 \\ \Gamma_3 L_v + \Gamma_4 N_v & \Gamma_3 L_p + \Gamma_4 N_p & \Gamma_3 L_r + \Gamma_4 N_r & 0 \\ \Gamma_4 L_v + \Gamma_8 N_v & \Gamma_4 L_p + \Gamma_8 N_p & \Gamma_4 L_r + \Gamma_8 N_r & 0 \\ 0 & 1 & \tan \theta_0 & 0 \end{pmatrix} \quad \mathbf{x}_{lat} = \begin{pmatrix} \Delta v \\ \Delta p \\ \Delta r \\ \Delta \phi \end{pmatrix}$$

$$\mathbf{B}_{lat} = \begin{bmatrix} \frac{Y_{\delta_a}}{m} & \frac{Y_{\delta_r}}{m} \\ \frac{L_{\delta_a}}{I'_x} + I'_{zx} N_{\delta_a} & \frac{L_{\delta_r}}{I'_x} + I'_{zx} N_{\delta_r} \\ I'_{zx} L_{\delta_a} + \frac{N_{\delta_a}}{I'_z} & I'_{zx} L_{\delta_r} + \frac{N_{\delta_r}}{I'_z} \\ 0 & 0 \end{bmatrix} \quad \mathbf{u}_{lat} = \begin{pmatrix} \Delta \delta_a \\ \Delta \delta_r \end{pmatrix}$$

Linearized Augmented lateral fixed-wing aircraft equations of motion for steady level flight:

$$\begin{pmatrix} \Delta \dot{v} \\ \Delta \dot{p} \\ \Delta \dot{r} \\ \Delta \dot{\phi} \\ \Delta \dot{\psi} \\ \Delta \dot{y}_E \end{pmatrix} = \left(\begin{array}{cccc|cc} & & & & 0 & 0 \\ & & & & 0 & 0 \\ & & & & 0 & 0 \\ & & & & 0 & 0 \\ \hline 0 & 0 & \sec \theta_0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & u_0 \cos \theta_0 & 0 \end{array} \right) \begin{pmatrix} \Delta v \\ \Delta p \\ \Delta r \\ \Delta \phi \\ \Delta \psi \\ \Delta y_E \end{pmatrix}$$

Lateral modal approximations:

Characteristic equation for $\mathbf{A}_{lat}$: $A\lambda^4 + B\lambda^3 + C\lambda^2 + D\lambda + E = 0$, where

$$E = g [(\mathcal{N}_r \mathcal{L}_v - \mathcal{N}_v \mathcal{L}_r) \cos \theta_0 + (\mathcal{N}_v \mathcal{L}_p - \mathcal{L}_v \mathcal{N}_p) \sin \theta_0]$$

$$D = -g (\mathcal{L}_v \cos \theta_0 + \mathcal{N}_v \sin \theta_0) + u_0 (\mathcal{L}_v \mathcal{N}_p - \mathcal{L}_p \mathcal{N}_v)$$

$$\mathbf{A}_{lat} = \begin{pmatrix} \mathcal{Y}_v & \mathcal{Y}_p & \mathcal{Y}_r & g \cos \theta_0 \\ \mathcal{L}_v & \mathcal{L}_p & \mathcal{L}_r & 0 \\ \mathcal{N}_v & \mathcal{N}_p & \mathcal{N}_r & 0 \\ 0 & 1 & \tan \theta_0 & 0 \end{pmatrix}$$

Longitudinal Dimensional Derivatives (Table 4.4 in Etkin and Reid):

	X	Z	M
u	$\rho u_0 SC_{w_0} \sin \theta_0 + \frac{1}{2} \rho u_0 SC_{x_u}$	$-\rho u_0 SC_{w_0} \cos \theta_0 + \frac{1}{2} \rho u_0 SC_{z_u}$	$\frac{1}{2} \rho u_0 \bar{c} SC_{m_u}$
w	$\frac{1}{2} \rho u_0 SC_{x_\alpha}$	$\frac{1}{2} \rho u_0 SC_{z_\alpha}$	$\frac{1}{2} \rho u_0 \bar{c} SC_{m_\alpha}$
q	$\frac{1}{4} \rho u_0 \bar{c} SC_{x_q}$	$\frac{1}{4} \rho u_0 \bar{c} SC_{z_q}$	$\frac{1}{4} \rho u_0 \bar{c}^2 SC_{m_q}$
$\dot{w}$	$\frac{1}{4} \rho \bar{c} SC_{x_{\dot{\alpha}}}$	$\frac{1}{4} \rho \bar{c} SC_{z_{\dot{\alpha}}}$	$\frac{1}{4} \rho \bar{c}^2 SC_{m_{\dot{\alpha}}}$

Lateral Dimensional Derivatives (Table 4.5 in Etkin and Reid):

	Y	L	N
v	$\frac{1}{2} \rho u_0 SC_{y_\beta}$	$\frac{1}{2} \rho u_0 b SC_{l_\beta}$	$\frac{1}{2} \rho u_0 b SC_{n_\beta}$
p	$\frac{1}{4} \rho u_0 b SC_{y_p}$	$\frac{1}{4} \rho u_0 b^2 SC_{l_p}$	$\frac{1}{4} \rho u_0 b^2 SC_{n_p}$
r	$\frac{1}{4} \rho u_0 b SC_{y_r}$	$\frac{1}{4} \rho u_0 b^2 SC_{l_r}$	$\frac{1}{4} \rho u_0 b^2 SC_{n_r}$